

Name : Parth Vishnu PRN: 24070521279

```
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO: ~ — ./server
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO:~/Desktop/CC LAB$ cd
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO:~$ hostname -I
10.17.60.51 10.17.26.230
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO:~$ vim server1279.c
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO:~$ gcc server1279.c -o server
sit-lab2-pc38@sit-lab2-pc38-OptiPlex-3280-AIO:~$ ./server
Server listening on port 10279...
Received Numbers: 5 2
Received Numbers: 3 4
```

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <netinet/in.h>

#define PORT 10000+279 // Replace 55 with your PRN last two digits
#define BUF_SIZE 1024

// Parameterized function
int add(int a, int b)
{
    return a + b;
}

int main()
{
    int server_fd, new_socket;
    struct sockaddr_in address;
    int addrlen = sizeof(address);

    char buffer[BUF_SIZE];
    int num1, num2, result;

    // Create socket
    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    // Server details
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);

    // Bind
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Listen
    listen(server_fd, 3);

    printf("Server listening on port %d...\n", PORT);

    // Accept connection
    new_socket = accept(server_fd, (struct sockaddr *)&address,
        (socklen_t *)&addrlen);

    while (1)
    {
        memset(buffer, 0, BUF_SIZE);

        int bytes = read(new_socket, buffer, BUF_SIZE);

        if (bytes <= 0)
            break;

        sscanf(buffer, "%d %d", &num1, &num2);
        printf("Received Numbers: %d %d\n", num1, num2);

        result = add(num1, num2);
        sprintf(buffer, "Sum = %d", result);

        send(new_socket, buffer, strlen(buffer), 0);
    }

    close(new_socket);
    close(server_fd);

    return 0;
}
```

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```
lab-03-44@lab-03-44-OptiPlex-3280-AIO:~$ ping 10.17.60.51
PING 10.17.60.51 (10.17.60.51) 56(84) bytes of data.
64 bytes from 10.17.60.51: icmp_seq=1 ttl=64 time=0.380 ms
64 bytes from 10.17.60.51: icmp_seq=2 ttl=64 time=0.212 ms
64 bytes from 10.17.60.51: icmp_seq=3 ttl=64 time=0.214 ms
64 bytes from 10.17.60.51: icmp_seq=4 ttl=64 time=0.545 ms
64 bytes from 10.17.60.51: icmp_seq=5 ttl=64 time=0.087 ms
^C
--- 10.17.60.51 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4101ms
rtt min/avg/max/mdev = 0.087/0.287/0.545/0.158 ms
lab-03-44@lab-03-44-OptiPlex-3280-AIO:~$ vim client1257.c
lab-03-44@lab-03-44-OptiPlex-3280-AIO:~$ gcc client1257.c -o client
lab-03-44@lab-03-44-OptiPlex-3280-AIO:~$ ./client
Enter two numbers: 5 2
Server Response: Sum = 7
Enter two numbers: 3
4
Server Response: Sum = 7
Enter two numbers:
```

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <netinet/in.h>

#define PORT 10000+279
#define BUF_SIZE 1024

int main()
{
    int sock;
    struct sockaddr_in server;
    char buffer[BUF_SIZE];
    int num1, num2;

    sock = socket(AF_INET, SOCK_STREAM, 0);

    server.sin_family = AF_INET;
    server.sin_port = htons(PORT);

    // Replace with the server PC's IP address
    server.sin_addr.s_addr = inet_addr("10.17.60.51");

    connect(sock, (struct sockaddr *)&server, sizeof(server));

    while (1)
    {
        printf("Enter two numbers: ");
        scanf("%d%d", &num1, &num2);

        sprintf(buffer, "%d %d", num1, num2);

        send(sock, buffer, strlen(buffer), 0);

        memset(buffer, 0, BUF_SIZE);

        read(sock, buffer, BUF_SIZE);

        printf("Server Response: %s\n", buffer);
    }

    close(sock);

    return 0;
}
```